

32 The Segregation and Maximum Mixedness Models

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Why Are We Learning This?

In the previous RTD lectures, we learned how to measure residence time distributions and how to characterize the mixing behavior of real reactors. However, knowing the RTD alone does not tell us the reactor conversion. We still need a way to connect reactor hydrodynamics to reactor performance.

For first-order reactions, RTDs are often sufficient because the conversion depends only on residence time. For non-first-order reactions, however, mixing becomes critically important. Two reactors may have the same RTD but different conversion if molecules mix differently inside the reactor.

To address this problem, engineers use mixing models. In this course, we focus on two important limiting cases:

- Segregation Model
- Maximum Mixedness Model

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



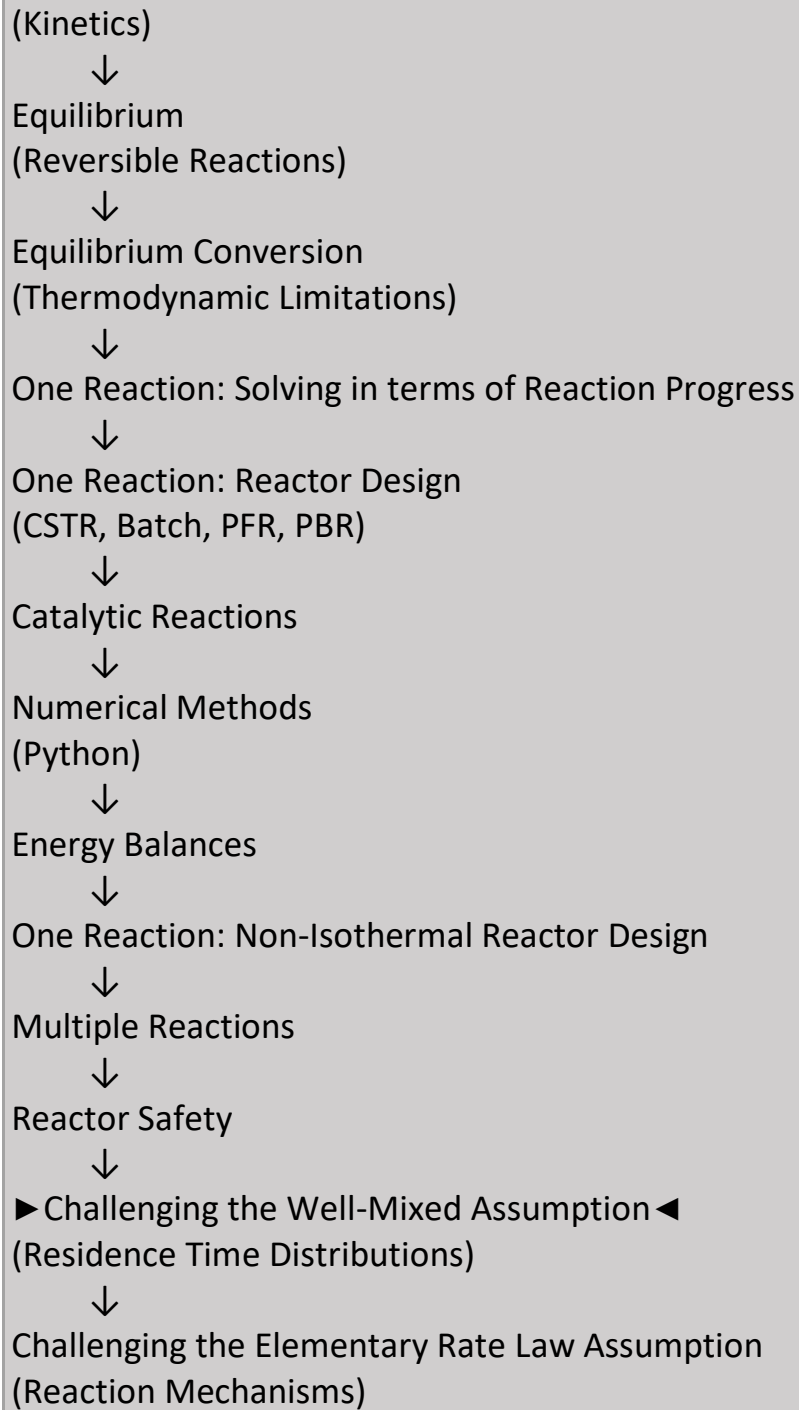
Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws



Learning Objectives

By the end of this lecture, you should be able to:

- Explain why RTDs alone are not sufficient for non-first-order reactions.
- Define molecular age and globules.
- Describe the assumptions behind the segregation model.
- Describe the assumptions behind the maximum mixedness model.
- Predict which model produces higher conversion for different reaction orders.
- Calculate mean conversion using the segregation model.
- Explain how the maximum mixedness model differs from segregation.

Big Idea

Residence Time Distribution

+

Mixing Model

↓

Predict Conversion

An RTD tells us how long molecules stay in a reactor. A mixing model tells us how those molecules interact while they are there. Both are needed to predict conversion in non-ideal reactors.

When we classify a reactor as ideal what we are saying is that the contents are

The degree of mixing is important, because it defines the
which defines the

The rate then determines the

Why Does Mixing Matter?

The degree of mixing affects:

Local Concentration

↓

Reaction Rate

↓

Conversion

Since most reaction rates depend on concentration, different mixing patterns can produce different conversions even when the residence time distribution is identical.

Exception: First-Order Reactions

For a first-order reaction: $\text{Rate} \propto \text{Concentration}$

The effects of mixing cancel out mathematically. As a result:

Conversion Depends Only on Residence Time and not on the details of mixing.

The exception is for $k_1 \gg k_2$, where the conversion does not depend on the concentration

Mole balances (in terms of conversion) for CSTR, PFR, and batch reactor for a first order reaction. Assuming constant volumetric flowrate:

- CSTR: $V = \frac{v_0 X}{k(1-X)}$
- PFR: $\frac{dX}{dV} = \frac{k(1-X)}{v_0}$
- Batch: $\frac{dX}{dt} = k(1 - X)$

Real reactors in general are not well mixed. Hence for non-first order reactions, we need methods for determining the concentration

Ideally, there would be a function to tell us this, but that is not reasonable. Instead, we use

These models differ in how they treat mixing amongst molecules of different "ages"

In mixing models, "age" means

What Is Molecular Age?

In RTD theory, **age** means:

Time Since a Molecule Entered the Reactor

Molecules having the same age are grouped together into: Globules

A globule is a conceptual packet of fluid whose molecules all entered the reactor at approximately the same time. Globules are used as a modeling tool to represent how fluid elements move and mix within a reactor.

Physical Interpretation

Enter Reactor Together



Remain in Same Globule



Share the Same Age

Each globule represents a fluid element that can be tracked as it moves through the reactor. Different globules may have different ages, depending on how long they have been inside the reactor.

Why Is This Idea Useful?

The segregation and maximum mixedness models are built around how globules of different ages interact:

Segregation Model



Globules Do Not Mix

Maximum Mixedness Model



Globules Mix Continuously

Therefore, understanding molecular age and globules is the foundation for understanding these reactor-mixing models.

"Groups" of molecules of the same age are called

We can draw a globule like

Globules within a reactor might look like (remember this is a model...)

There are a variety of ways to model mixing in reactors. In this class, we will focus on the

- Segregation model: Globules do not mix with other globules until

- Maximum mixedness model: Globules mix with other globules as soon as they

In this model, the fluid is completely mixed at the

Segregation vs Maximum Mixedness

These models represent two opposite limits of mixing behavior.

Segregation Model: Globules Remain Separate

Different-age fluid elements do not mix with one another until they leave the reactor.

Maximum Mixedness Model: Globules Mix Immediately

Fluid elements continually mix with other fluid elements as soon as they enter the reactor.

Comparison

Segregation



No Interactions Between Ages

Maximum Mixedness



Maximum Interaction Between Ages

Real reactors generally behave somewhere between these two extremes.

For single reactions with orders greater than one or less than zero, the segregation model will predict the highest conversion. For reaction orders between zero and one, the maximum mixedness model will predict the highest conversion.

For reactions with orders of one

Why Does Reaction Order Matter?

The effect of mixing depends strongly on the reaction order.

Reaction Orders Greater Than One: Higher concentrations produce much higher reaction rates.

Therefore: Keeping Molecules Segregated Often Preserves High Concentration Regions leading to higher conversion.

As a result: Segregation Model Usually Predicts Highest Conversion for orders greater than one.

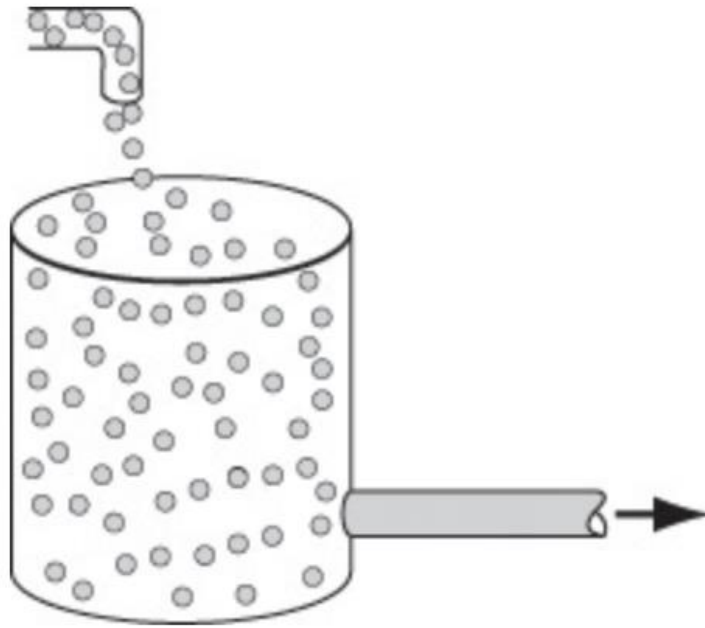
Reaction Orders Between Zero and One

In this range: Mixing Becomes Beneficial and the maximum mixedness model tends to predict higher conversion.

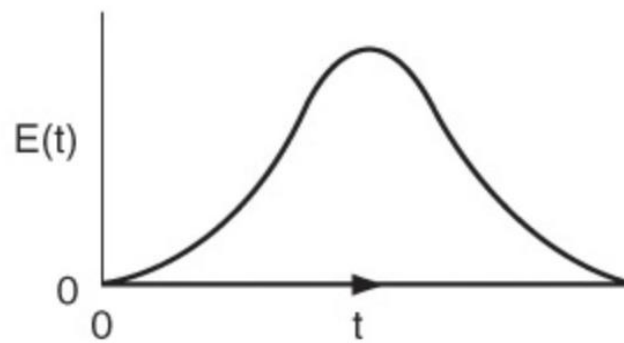
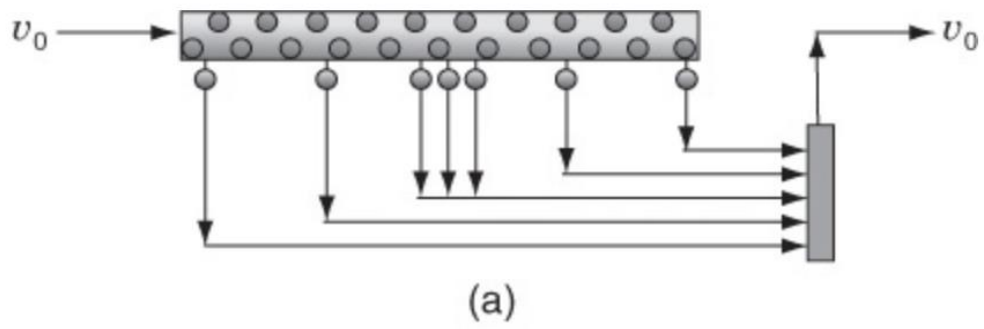
First-Order Reactions

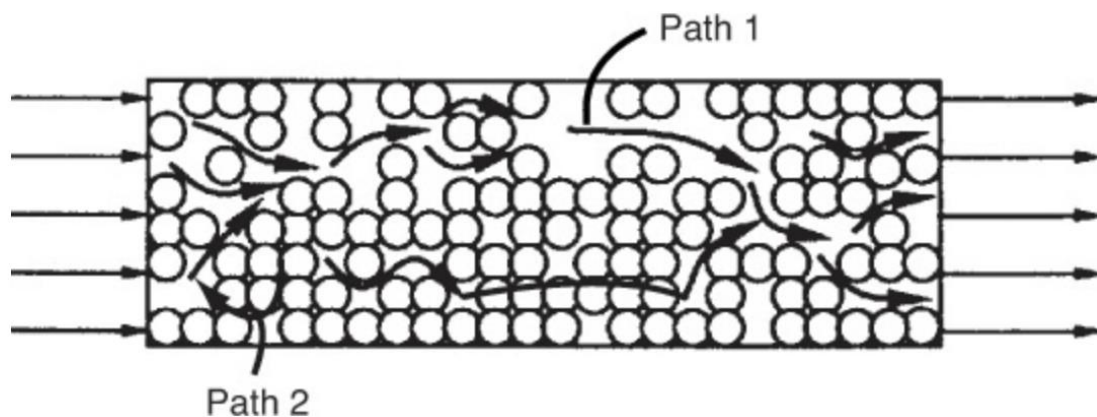
For Reaction Order = 1: Both models give the same result.

- Segregation model in a CSTR



- Segregation model in a PFR





In the segregation model, there is no mixing between globules, and hence each globule can be modeled as a

Why Is the Segregation Model Easier?

Under the segregation assumption: No Mixing Between Globules. Each globule behaves independently.

Therefore: Each Globule Can Be Treated As A Tiny Batch Reactor.

The conversion achieved by a globule depends only on: Its Residence Time.

This greatly simplifies the analysis.

Physical Interpretation

Globule

↓

Miniature Batch Reactor

↓

Compute Conversion

↓

Weight by RTD

The mean conversion is given by:

Mean conversion of globules spending between time t and $t + \Delta t$ in the reactor = Conversion achieved in a globule spending time t in the reactor $\times$ Fraction of globules that spend between time t and $t + \Delta t$ in the reactor. Mathematically, this looks like

Rearranging: $\frac{d\bar{X}}{dt} = X(t)E(t)$. Integrating:

Thus, if we have the $\bar{X}$ and the $E(t)$, we can find the mean conversion.

Most Important Result of the Segregation Model

To predict the mean conversion, we need:

1. The conversion achieved by a globule as a function of time, $X(t)$.
2. The residence time distribution, $E(t)$.

The overall reactor conversion is then obtained as a weighted average: Conversion of a Globule $\times$ Fraction of Globules Having That Residence Time

In other words, globules that spend more time in the reactor generally achieve higher conversion, while the RTD determines what fraction of the fluid experiences each residence time.

Workflow

Solve Batch Reactor



Obtain $X(t)$



Combine with $E(t)$



Calculate Mean Conversion

Why This Is Powerful

Under the segregation model, each globule behaves independently and can be treated as a tiny batch reactor. This allows us to use familiar batch-reactor design equations together with the measured RTD to predict the performance of a non-ideal reactor.

Key Insight

RTD



Tells How Long Globules Stay

$X(t)$



Tells What Conversion Each Globule Achieves

Combine Both

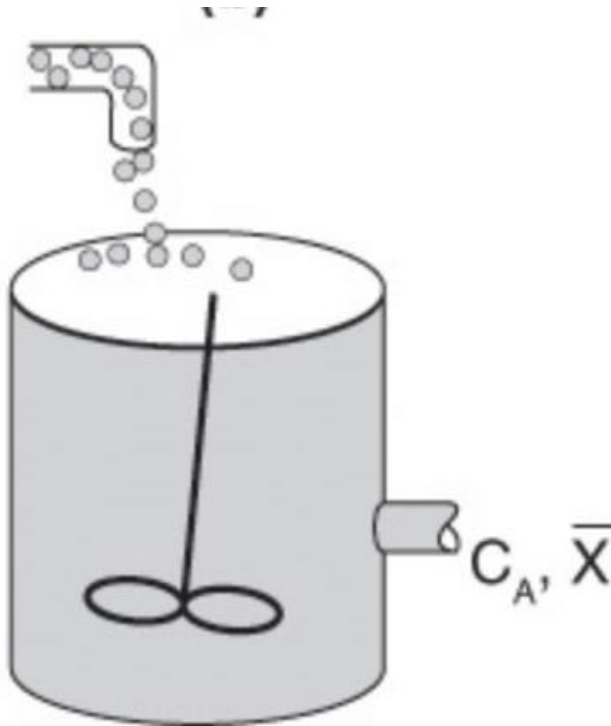


Predict Reactor Conversion

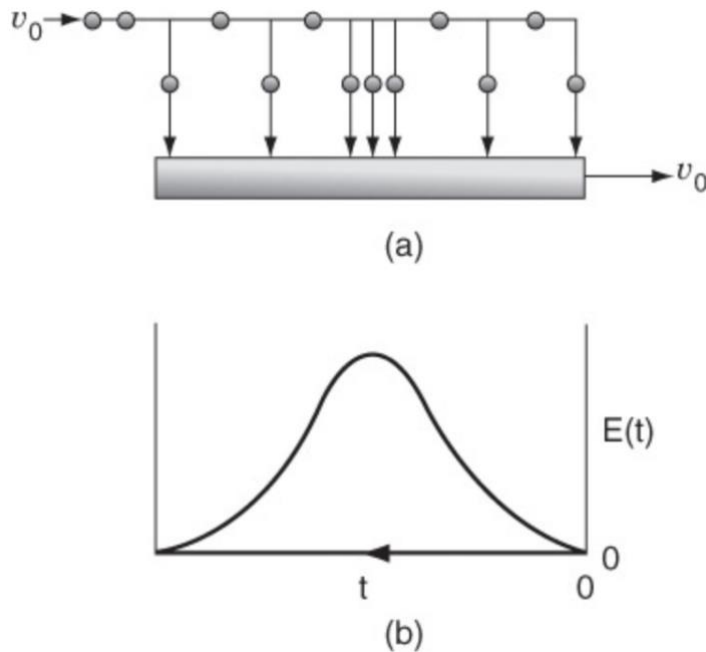
The segregation model converts a difficult non-ideal reactor problem into two simpler problems: determining the RTD and solving the batch-reactor conversion as a function of time.

In the maximum mixedness model, molecules mix

- Maximum mixedness model in a CSTR



- Maximum mixedness model in a PFR



The derivation for the mean conversion under the maximum mixedness model involves a lot of math that I am going to skip. But you can find it [here](#). Basically, we define a new variable λ , which is the time it takes a molecule to move to the end of the reactor.

We then do a mole balance on species A in this reactor model, and arrive at $\frac{dX}{dz} = -\frac{r_A}{C_{A0}} - \frac{E(\bar{T}-z)}{1-F(\bar{T}-z)}X$, where z is the difference between the longest time measured in the $E(t)$ curve, $\bar{T}$ and λ . I.e., $\lambda = \bar{T} - z$.

Why Is the Maximum Mixedness Model More Difficult?

The segregation model is relatively simple because each globule behaves independently. The maximum mixedness model is more difficult because molecules continuously mix with molecules of different ages.

Physical Interpretation

Young Molecules

↓
Mix With
↓
Older Molecules
↓
Concentrations Change Continuously

Unlike the segregation model, a molecule's environment is constantly changing as mixing occurs throughout the reactor.

Consequence

As a result, Conversion Depends On: Residence Time + Mixing History

Both how long a molecule remains in the reactor and how it interacts with other molecules influence the overall conversion.

Why the Mathematics Is More Complex

In the segregation model: Each Globule Can Be Solved Independently

In the maximum mixedness model: All Globules Interact Through Mixing

Because the behavior of one globule depends on the behavior of other globules, the mathematical treatment is much more complicated and typically requires solving a differential equation involving both residence time and mixing.

Key Insight

Segregation
↓
Independent Globules
↓
Simpler Mathematics

Maximum Mixedness
↓
Interacting Globules
↓
More Complex Mathematics

The additional complexity arises because molecular mixing and reaction are occurring simultaneously throughout the reactor.

Summary and Key Takeaways

Today we introduced mixing models used to connect RTDs to reactor conversion.

Key Ideas

- ✓ RTDs alone are not sufficient for non-first-order reactions.
- ✓ Mixing influences concentration, which influences reaction rate.
- ✓ Molecules of the same age are grouped into globules.
- ✓ The segregation model assumes no mixing between globules.
- ✓ The maximum mixedness model assumes immediate mixing between globules.
- ✓ Real reactors typically fall between these two extremes.
- ✓ For first-order reactions, both models predict the same conversion.
- ✓ For non-first-order reactions, the two models can produce significantly different predictions.

Most Important Idea

Residence Time Distribution



Tells How Long Molecules Stay

Mixing Model



Tells How Molecules Interact

Both Together



Determine Conversion

The central lesson in this lecture is that residence time alone does not determine reactor performance. Mixing behavior must also be considered whenever reaction rates depend nonlinearly on concentration.